

Predicting Breastfeeding Duration Using the LATCH Breastfeeding Assessment Tool

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Abstract

The authors tested the validity of the LATCH breastfeeding assessment tool, controlling for intervening variables in 133 dyads. LATCH scores, mother's evaluation of an index feed, and intended duration of breastfeeding were assessed postpartum and followed 6 weeks. Women breastfeeding at 6 weeks postpartum had higher LATCH scores ($\bar{x} \pm SD = 9.3 \pm 0.9$) than those who weaned ($\bar{x} \pm SD = 8.7 \pm 1.0$), due to only one measure, breast/nipple comfort. Women who weaned before 6 weeks reported lower breast/nipple comfort (1.5 ± 0.5) than those who were still breastfeeding at 6 weeks (1.7 ± 0.5 , $P < .05$). Total LATCH scores accounted for 7.3% of variance in breastfeeding duration. Total LATCH scores positively correlated with duration of breastfeeding ($n = 128$; $r = .26$, $P = .003$) and to mothers' scores ($n = 132$; $r = .58$, $P = .001$). Correlations among LATCH measures ranged from .02 to .51. The LATCH tool is a useful identifies the need for follow-up with breastfeeding mothers at risk for early weaning because of sore nipples. *J Hum Lact* 2001;17(1):20-23.

Keywords: LATCH, breastfeeding, assessment tool, predictor, duration

A short clinical assessment tool that scores early breastfeedings is needed to identify mothers whose infants are at risk for early weaning. Such a risk assessment tool would help postpartum nurses focus on those mothers and their babies who need special assistance in the hospital and would provide data that justify close follow-up after discharge.¹ In addition, nurses could use a valid breastfeeding assessment tool to improve their ability to document assessments, measure quality of care, and conduct research.

Breastfeeding is not a single behavior but a series of behaviors. Several breastfeeding assessment tools have tried to tap into the domains of feedings at the breast. One of these tools, LATCH,² is used widely by nurses at hospitals and birth centers. The LATCH tool, which is based on observations and descriptions of effective breastfeeding, evaluates five characteristics of breastfeeding. A numerical score (0, 1, or 2) is assigned to each measure for a possible total score of 10. Each letter of the acronym denotes a category. *L* represents how well the infant latches onto the breast, *A* represents audible swallowing noted, *T* describes the mother's nipple type, *C* represents the mother's degree of breast or nipple comfort, and *H* evaluates the amount of help the mother needs to position her baby at breast. Because a tool must be valid to be useful, this study examined the validity of the LATCH tool by comparing it with other measures of effective breastfeeding and by determining its effectiveness in predicting breastfeeding duration to 8 weeks postpartum.

Method

The study took place in two Midwestern US community hospitals. One of the hospitals is located in a university town, the other in a medium-sized city. Lactation

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consultants assessed feedings between April 1995 and May 1996. The average postpartum stay was 3 to 4 days, compared with the current average of 2 to 3 days. Feedings were evaluated using the LATCH tool 24 to 72 hours postpartum. At each hospital, breastfeeding was assessed by nurses (certified as lactation consultants) who observed the mother and infant in the mother's quiet room while conducting daily rounds on an as-they-come basis. The sample excluded infants who were admitted to the neonatal intensive care unit, were born less than 38 or more than 42 weeks, or were twins. No eligible mothers refused to participate. Mother-infant couplets were followed 8 weeks postpartum.

At the time of the assessment, each mother was asked to rate how she thought the breastfeeding went (0 = *not at all well* to 3 = *very well*) and how long she intended to breastfeed this baby using four categories: 2 weeks, 4 weeks, 6 weeks, or 2 months or more. Information about the baby's birth, the mother's age, and previous breastfeeding experience was also collected. An analysis of variance showed no significant differences in mean LATCH scores between nurse evaluators ($P = .16$).

At 8 weeks postpartum, each mother was called and asked the mode of infant feeding. "Not breastfeeding" was defined as not having breastfed in the previous 24 hours and not intending to breastfeed this baby further. Actual length of breastfeeding and the mother's intended breastfeeding duration was categorized as less than 2 weeks, 2 to 4 weeks, 4 to 6 weeks, or more than 6 weeks. Statistical analyses were conducted using StatView version 5.0.1. (SAS Institute, Inc.).

Analysis of variance was conducted to identify differences in LATCH scores between groups still breastfeeding at 6 weeks postpartum and those who had weaned, controlling for demographic factors. Stepwise multiple regression examined the relationship between LATCH scores and actual breastfeeding duration. Spearman correlations were conducted to determine relationships between LATCH scores and clinical and demographic variables. Descriptive statistics were calculated for the LATCH tool and demographic data.

Results

The average infant in the sample weighed 3535 ± 468 g ($\bar{x} \pm SD$) at birth, 17% were born by cesarean, and half were males. Mothers were an average of 29 ± 5 years old. On average, study infants were 39 ± 14 hours old when the assessment took place. The majority evaluated

Table 1. Participant Characteristics (N = 127)

	Breastfeeding at 6 wks, n = 90	Not Breastfeeding at 6 wks, n = 37
First-time breaster, n (%)	44 (48)	22 (59)
Cesarean birth, n (%)	13 (14)	9 (25)
Infant gender, % male	49	51
Birth weight, g ($\bar{x} \pm SD$)	3543 ± 440	3518 ± 559
Mother's age, y ($\bar{x} \pm SD$)	29 ± 5	27 ± 5
Assessment age, hr ($\bar{x} \pm SD$)	40 ± 14	37 ± 17
Total LATCH score ($\bar{x} \pm SD$)	9.3 ± 0.9	$8.7 \pm 1.0^*$
Latch-on	1.9 ± 0.4	1.8 ± 0.4
Audible swallowing	1.9 ± 0.4	1.7 ± 0.5
Nipple type	1.9 ± 0.2	1.9 ± 0.3
Comfort of breast/nipple	1.7 ± 0.5	$1.5 \pm 0.5^*$
Help needed to position baby	1.9 ± 0.3	1.8 ± 0.3

* $P < .05$, Fisher's exact test.

the index breastfeeding as 3 = *pleased* (30%) or 4 = *very pleased* (58%). All participants indicated that they intended to breastfeed for at least 6 weeks (20%) or for at least 8 weeks or more (80%). Although most mothers (71%) were still breastfeeding at 6 weeks postpartum, 10% had weaned by 2 weeks. At 8 weeks postpartum, almost one half of the women reported that their infants had received one or more feedings other than breast milk, mostly formula. The average LATCH score was 9.0 ± 1.0 and ranged from 5 to 10.

Table 1 compares subject characteristics between women still breastfeeding at 6 weeks postpartum and those who had weaned. The percentage of women breastfeeding for the first time, type of delivery, infant birth weight and gender, and mother's age did not differ significantly nor did the age the assessment took place postpartum.

Women who were breastfeeding at 6 weeks postpartum had higher total LATCH scores (9.3 ± 0.9) than those who had weaned (8.7 ± 1.0) ($P < .05$). All means of LATCH measures (latch-on, audible swallowing, nipple type, maternal comfort, and help needed in getting the baby to the breast) tended to be higher in the group still breastfeeding at 6 weeks; however, the differences were not significant, except for comfort of breast/nipple (sore nipples). Women who weaned before 6 weeks reported lower breast/nipple comfort (1.5 ± 0.5) than those who were still breastfeeding at 6 weeks (1.7 ± 0.5), ($P < .05$) (Table 1).

Stepwise multiple regression analysis was used to examine the variables associated with breastfeeding duration. Total LATCH score, mother's age, intended

Table 2. Correlation Coefficients for Association Between Breastfeeding Duration With LATCH Subscores

	<i>Mothers' Score</i> (<i>n</i> = 132)	<i>Latch-on</i> (<i>n</i> = 131)	<i>Audible Swallow</i> (<i>n</i> = 131)	<i>Nipple Type</i> (<i>n</i> = 131)	<i>Comfort of Breast/Nipple</i> (<i>n</i> = 131)	<i>Help needed</i> (<i>n</i> = 130)	<i>Duration</i> (<i>n</i> = 128)
Total score	.58 [†]	.69 [†]	.66 [†]	.33 [†]	.36 [†]	.52 [†]	.26 [*]
Mothers' score		.40 [†]	.38 [†]	.16	.17	.38 [†]	.22 [*]
Latch-on			.51 [†]	.13	-.08	.26 ^{**}	.09
Audible swallowing				.02	-.18 [*]	.25 ^{**}	.10
Nipple type					.06	.25 ^{**}	.07
Comfort						.01	.20 [*]
Help needed							.17

* $P < .05$.** $P < .01$.† $P < .001$.

breastfeeding duration, and type of delivery were all significantly related ($R^2 = .25$, $P = .006$). Previous breastfeeding experience, infant gender, and birth weight did not enter the equation. The total LATCH score accounted for 7.3% of the variance in length of breastfeeding, followed by mother's age (6.7%), type of delivery (5.1%), and maternal intent (5.1%).

Construct validity was assessed by conducting Spearman correlations between selected variables (Table 2). Mothers' scores were significantly correlated with the evaluators' scores ($n = 132$; $r = .56$, $P = .001$). In addition, mothers' score ($n = 127$; $r = .22$, $P = .012$) and the total LATCH score were positively correlated with duration of breastfeeding ($n = 128$; $r = .26$, $P = .003$).

Since the LATCH tool should cover five distinct dimensions of breastfeeding, further empirical testing of the construct validity of the LATCH tool was accomplished by calculating Spearman correlations among LATCH measures. It was expected that the magnitude of the correlations would be low because the individual items are theoretically independent of each other. Table 2 displays the correlations, most of which were low, with a range of .02 for audible swallowing with nipple type to .26 for help needed with latch-on. Contrary to expectations, however, the correlation between latch-on and audible swallowing was moderately high (.51), indicating a lack of independence.

Discussion

This study assessed the performance of the LATCH tool score of an index feed to see if it predicted whether the mother would be breastfeeding at 6 weeks, taking into consideration the mother's planned breastfeeding goals and other demographic variables. Six weeks was chosen as the outcome variable because it is a vulnera-

ble time when the mother's milk supply is being established and when both mother and baby are learning the skill of breastfeeding.

Reports of the LATCH tool's validity have been mixed. Riordan and Koehn³ found that correlations between the total LATCH score and two other breastfeeding assessment tool scores were relatively low (.11-.48). However, this study tends to support the validity of the LATCH. Mothers' feeding assessment scores were similar to those of the nurse evaluators, and both were positively related to breastfeeding duration. The similarity between the mothers' scores and evaluators' scores found in this study is consistent with previous research. Both Adams and Hewell⁴ and Matthews⁵ found an association between the mother's satisfaction with the feed and the evaluator's score. Schlomer et al.⁶ compared LATCH scores with maternal satisfaction and breastfeeding problems. As LATCH scores increased, maternal satisfaction tended to increase and breastfeeding problems tended to decrease.

LATCH scores predicted whether or not the mother would breastfeed for up to 6 weeks; thus, the LATCH tool would aid nurses in identifying those who potentially will stop breastfeeding prematurely and who might need interventions to continue to breastfeed. Mothers who scored lower on comfort (they were engorged or had cracked or bleeding nipples and were in severe discomfort) were less likely to be breastfeeding at 6 weeks postpartum, despite their earlier intentions to feed at least that long.

The relative independence between various dimensions of the LATCH measure except between latch-on and audible swallowing suggests that the LATCH tool measures separate domains of feedings at the breast. Latch-on and audible swallowing could be combined

into one variable since they overlap. Audible swallowing has another disadvantage in that it is time dependent and is not a viable measure until the third or fourth day postpartum when the mother's milk comes in. It is interesting to note that the mother's need for help is significantly associated with all other measures except for breast/nipple comfort. Nipple type is independent of other measures except for help needed, probably reflecting that the mother with flat or inverted nipples needs special help in getting the baby to breast.

This study took place in the hospital setting; however, the LATCH tool would also be useful in community-based health care settings where mothers and infants are seen beyond the first few days postpartum. When the LATCH tool is used, we recommend that health professionals closely follow mothers who score 2.0 or lower on the nipple comfort measure, to avoid unintended early weaning and assist the mothers to reach their breastfeeding goals.

Further testing of construct validity is warranted. In particular, correlations between the amount of actual intake of breast milk and individual measures of the tool should be done. This study was limited in that assessment data were collected at one random feeding time, a cross-sectional snapshot of one feeding, not a composite score of several feedings that might better predict breastfeeding problems. In addition, for greater flexibility in data analysis, we suggest asking a mother how long she intends to breastfeed as an open-ended question rather than as one with a few responses (2, 4, 6, and 8 weeks) as used in this study. The LATCH tool is useful in identifying with breastfeeding mothers at risk for early weaning because of sore nipples.

References

1. Riordan J. Predicting breastfeeding problems. *AWHONN Lifelines*. December 1998;31-33.
2. Jensen D, Wallace S, Kelsay P. LATCH: a breastfeeding charting system and documentation tool. *J Obstet Gyn Neonat Nurs*. 1994;23:27-32.
3. Riordan J, Koehn M. Reliability and validity testing of three breastfeeding assessment tools. *J Obstet Gyn Neonat Nurs*. 1997;26:181-187.
4. Adams D, Hewell SD. Maternal and professional assessment of breastfeeding. *J Hum Lact*. 1997;13:279-283.
5. Matthews MK. Developing an instrument to assess infant breastfeeding behaviour in the early neonatal period. *Midwifery*. 1988;4:154-165.
6. Schlomer JA, Kemmerer J, Twiss JJ. Evaluating the association of two breastfeeding assessment tools with breastfeeding problems and breastfeeding satisfaction. *J Hum Lact*. 1999;15:35-39.

Resumen

Este estudio validó el instrumento LATCH de evaluación de la mamada, controlando variables entre 133 parejas madre-niño/a. Los puntajes de LATCH, el índice de evaluación de la mamada, y la intención de la duración de la lactancia se evaluaron en el postparto y a las 6 semanas. Las mujeres amamantando a las 6 semanas postparto tenían puntajes mayores ($\bar{x} \pm SD = 9.3 \pm 0.9$) que las que suspendieron la lactancia (8.7 ± 1.0), en un solo aspecto, alivio del pecho/areola. Las mujeres que suspendieron la lactancia antes de las 6 semanas reportaron menor alivio del pecho/areola (1.5 ± 0.5) comparado con otros puntajes altos (1.7 ± 0.5 , $P < .05$). El total del puntaje de LATCH fue útil en 7.3% en relación a la duración de la lactancia ($n = 128$; $r = .26$, $P = .003$) y en los puntajes de las madres ($n = 132$; $r = .56$, $P = .001$). El rango de medidas del LATCH fue entre .02 a .51. El instrumento de LATCH es útil para identificar la necesidad de seguir de cerca a las madres con un alto riesgo de suspender la lactancia en sus etapas tempranas.